Wolff–Kishner reduction

An aldehyde or ketone reacts with hydrazine to give a hydrazone. In the presence of a base, the reaction continues to form an alkane.

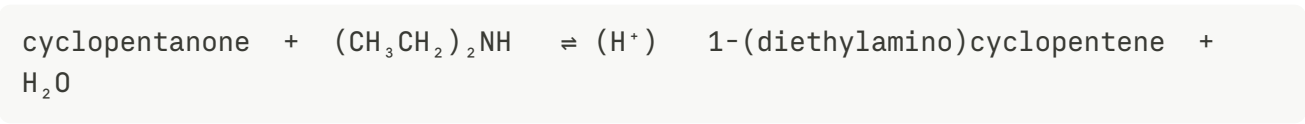


Mechanism:

1. Hydroxide removes an N–H proton of the hydrazone, giving R(R′)C=N–N⁻–H ↔ R(R′)C⁻–N=N–H.
2. The carbanion is protonated by water: R(R′)CH–N=N–H (+ OH^-).
3. Hydroxide removes the remaining N–H proton: R(R′)CH–N=N^- (+ H_2O).
4. Loss of N_2 gives the carbanion R(R′)CH^-, which is protonated by water: R–CH_2–R′ + OH^- (+ N_2).

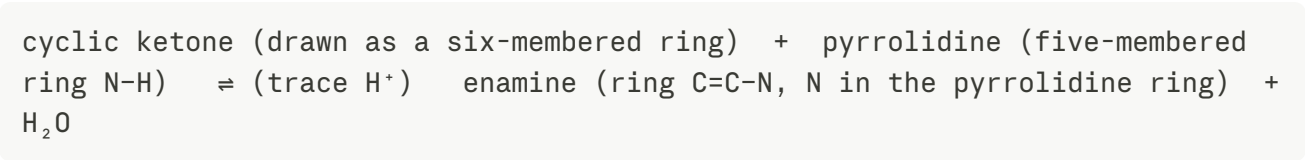
Enamines

Aldehydes and ketones react with secondary (2°) amines to form enamines.



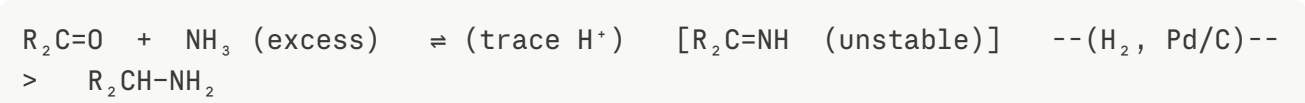
The mechanism is similar to that of imine formation: nucleophilic attack of the amine, proton transfers to the carbinolamine, protonation of –OH and loss of water give the iminium ion R_2N^+=C. **The intermediate cannot lose a proton from N, so it loses an H^+ from an α-carbon** (to B:), which gives the C=C of the enamine.

Exercise in the notes: propose a mechanism for



Reductive amination

An imine formed from the reaction of an aldehyde or ketone with NH_3 is relatively unstable. It serves as a useful intermediate. If the reaction with ammonia is carried out in the presence of H_2 and an appropriate metal catalyst, hydrogen adds to the C=N bond as it is formed, giving a primary amine. This reaction is called **reductive amination**.



Secondary and tertiary amines can be prepared from imines and enamines using sodium cyanoborohydride ($NaBH_3CN$). This is called the **Borch reaction**.

```
cyclohexanone ring C=O  +  R–NH2   ⇌ (trace H+)   ring C=N–R (imine)   --
(NaBH3CN)-->   ring CH–NHR   (2° amine)

ring C=O  +  R2NH   ⇌ (trace H+)   ring C=C–NR2 (enamine, drawn as ring–NR2)
--(NaBH3CN)-->   ring CH–NR2   (3° amine)
```

5. Reaction with water

Aldehydes and ketones react with water to form **hydrates (gem-diols)**.

```
R–C(=O)–R(H)  +  H2O   ⇌ (HCl)   R–C(OH)2–R(H)
```

Mechanism (acid-catalysed):

1. $H_2O + H^+ \rightleftharpoons H_3O^+$, and the carbonyl oxygen is protonated: $R–C(=O^+H)–R'$.
2. Water attacks the carbonyl carbon: $R–C(OH)(O^+H_2)–R'$.
3. Deprotonation by water gives the hydrate $R–C(OH)_2–R' + H_3O^+$.

Reactions of aldehydes and ketones with alcohols

Addition of 1 equivalent of alcohol to an aldehyde yields a **hemiacetal**. Addition of a 2nd equivalent of alcohol yields an **acetal**. Like water, an alcohol is a poor nucleophile, so an acid catalyst is required for the reaction to take place at a reasonable rate.

```
R–C(=O)–H  +  CH3OH   ⇌ (HCl)   R–CH(OH)(OCH3)   ⇌ (CH3OH, HCl)   R–CH(OCH3)2
+  H2O
 aldehyde                          hemiacetal                          acetal
```

In the case of ketones, the addition products are called a **hemiketal** and a **ketal** respectively:

```
R–C(=O)–R  +  CH3OH   ⇌ (HCl)   R–C(OH)(OCH3)–R   ⇌ (CH3OH, HCl)   R–
C(OCH3)2–R
                                   hemiketal                           ketal
```

Mechanism:

1. The carbonyl oxygen is protonated ($H–B^+$).
2. CH_3OH attacks the carbonyl carbon.

3. Deprotonation gives R–C(OH)(OCH_3)–R′: a **hemiacetal** if R′ = H, a **hemiketal** if R′ = alkyl.
4. The –OH is protonated and water leaves, giving R–C(=O^+CH_3)–R′.
5. A second CH_3OH attacks, and deprotonation gives R–C(OCH_3)$_2$–R′: an **acetal** if R′ = H, a **ketal** if R′ = alkyl.

The Wittig reaction

An aldehyde or a ketone reacts with a phosphonium ylide to form an alkene.

```
R(R')C=O  +  (C6H5)3P=CH-CH3   -->   R(R')C=CH-CH3  +  (C6H5)3P=O
             phosphonium ylide                          triphenylphosphine
oxide
             (R' = H, alkyl)
```

Mechanism: the ylide carbon attacks the carbonyl carbon (O^- and $P^+(C_6H_5)_3$ on adjacent atoms, a four-membered O–P ring intermediate), then the ring collapses to give the alkene and $O{=}P(C_6H_5)_3$.

Examples:

```
(CH3CH2)2C=O  +  (C6H5)3P=CH-CH2CH3   -->   (CH3CH2)2C=CH-CH2CH3  (3-
ethylhex-3-ene)  +  (C6H5)3P=O

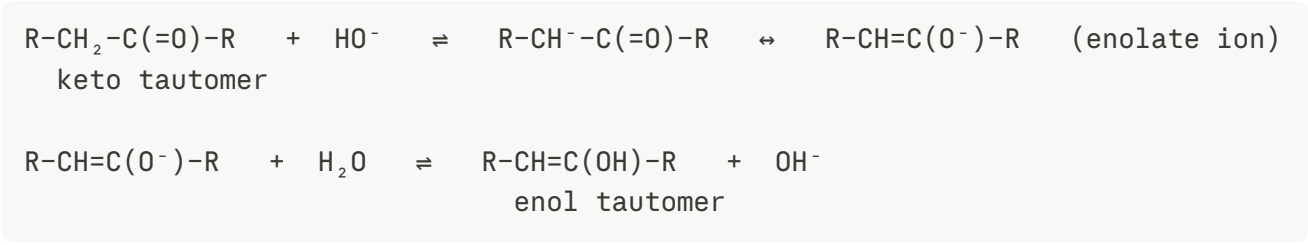
cyclic ketone (ring as drawn)  +  (C6H5)3P=CH2   -->   ring C=CH2  +
(C6H5)3P=O
```

6. Reactions at the α-carbon

Aldehydes and ketones have a second site for reactivity, i.e. the carbon bonded directly to the carbonyl carbon (the **α-carbon**). The hydrogen attached to the α-carbon (the **α-hydrogen**) is acidic enough to be removed by a strong base (the notes read "difficiently acidic", which I take to mean "sufficiently acidic").

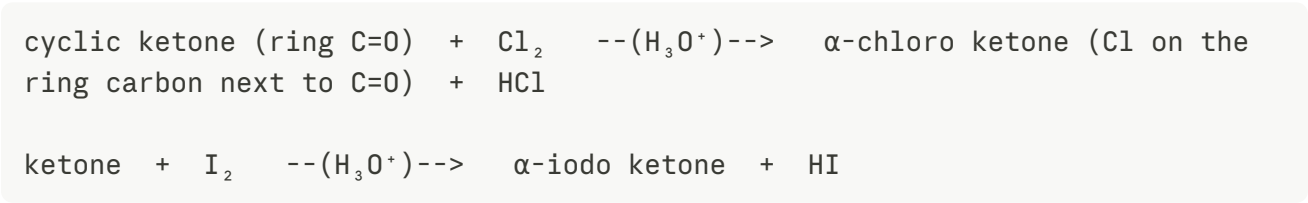
```
      H   O                        O
      |   ||                       ||
R – C – C – R   +  :B   ⇌   R – C⁻H – C – R   +  H–B
```

Keto–enol interconversion: enolate formation

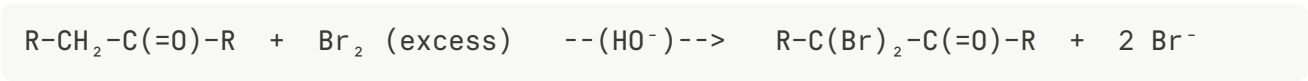


Halogenation of the α-carbon of aldehydes and ketones

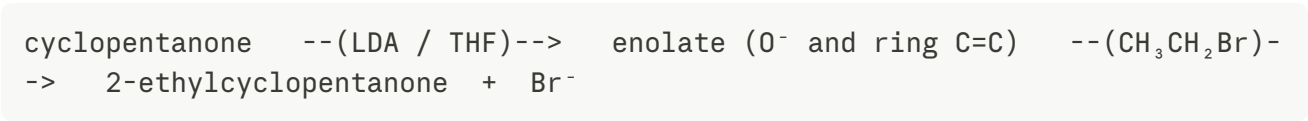
Acid-catalysed halogenation (the notes say: no need for the mechanism)



Base-promoted halogenation (marked "Not needed" in the notes)



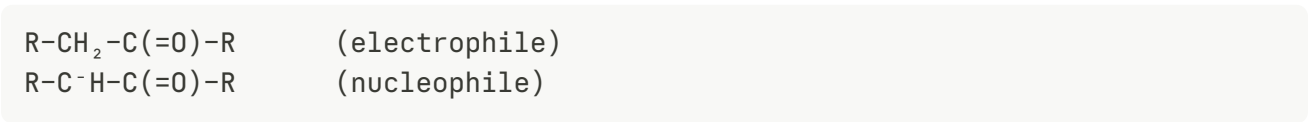
Alkylation of the α-carbon of carbonyl compounds



LDA = lithium diisopropylamide, $(CH_3)_2CH–N^-(Li^+)–CH(CH_3)_2$.

The aldol reaction

An **aldol addition** is a reaction in which one molecule of a carbonyl compound, after a proton is removed from an α-carbon, reacts as a nucleophile and adds to the electrophilic carbonyl carbon of a second carbonyl compound.



Examples:

$2\ CH_3CH_2–CHO$	$\rightleftharpoons (HO^-, H_2O)$	$CH_3CH_2–CH(OH)–CH(CH_3)–CHO$	β-hydroxyaldehyde
$2\ CH_3–CO–CH_3$	$\rightleftharpoons (HO^-, H_2O)$	$CH_3–C(OH)(CH_3)–CH_2–CO–CH_3$	β-hydroxyketone

Mechanism (aldehyde):

1. HO^- removes an α-H: $R–CH_2–CHO + HO^- \rightleftharpoons R–CH^-–CHO \leftrightarrow R–CH=CH–O^- + H_2O$
2. The enolate attacks the carbonyl carbon of a second aldehyde molecule: $R–CH_2–CH(O^-)–CHR–CHO$
3. Protonation by water gives the β-hydroxyaldehyde $R–CH_2–CH(OH)–CHR–CHO$ (and HO^- is regenerated).

Mechanism (acetone):

1. HO^- removes an α-H: $CH_3–CO–CH_3 \rightleftharpoons CH_3–CO–CH_2^- \leftrightarrow CH_3–C(O^-)=CH_2$
2. The enolate attacks a second acetone: $CH_3–CO–CH_2–C(CH_3)_2–O^-$
3. Protonation by water gives $CH_3–CO–CH_2–C(OH)(CH_3)_2$ (β-hydroxyketone) + HO^-.

Problems in the notes

1. Show the aldol addition product that would be formed from each of the following compounds:
   - (i) $CH_3CH_2–CO–CH_2CH_3$
   - (ii) cyclopentanone
   - (iii) $CH_3CH_2CH_2CH_2–CHO$
2. For each of the following compounds, indicate the aldehyde or ketone from which it would be formed by an aldol addition:
   - (i) 4-hydroxy-4-methyl-2-pentanone
   - (ii) 2,4-dicyclohexyl-3-hydroxybutanal

Aldol condensation

If the product of an aldol addition undergoes dehydration, the overall reaction is called an **aldol condensation**. An aldol condensation yields α,β-unsaturated carbonyl compounds.

```
R–CH2–CHO   ⇌ (HO⁻, H2O)   R–CH2–CH(OH)–CH(R)–CHO   --(H3O⁺, Δ)-->   R–CH2–CH=C(R)–CHO
                                                                                α,β-unsaturated compound

2 CH3–CO–CH3   ⇌ (HO⁻, H2O)   CH3–C(OH)(CH3)–CH2–CO–CH3   --(OH⁻, Δ)-->   (CH3)2C=CH–CO–CH3  +  H2O
```

Mechanism of the dehydration:

1. A base (B:) removes the α-H that lies between the carbinol carbon and the C=O: R–CH(OH)–CH(H)–CHO.
2. This gives the enolate R–CH(OH)–CH=C(O^-)H.
3. The enolate expels OH^- from the β-carbon: R–CH=CH–CHO.

The crossed aldol addition

If two different carbonyl compounds are used in an aldol reaction, **four products** are obtained. Example: propanal (A) + ethanal (B).

```
A = CH3CH2–CHO          B = CH3–CHO
A⁻ (enolate of A) = CH3–C⁻H–CHO          B⁻ (enolate of B) = ⁻CH2–CHO
(both formed with OH⁻, H2O)
```

Enolate	Attacks	Product
A^-	A	① CH_3CH_2–CH(OH)–CH(CH_3)–CHO
A^-	B	② CH_3–CH(OH)–CH(CH_3)–CHO
B^-	A	③ CH_3CH_2–CH(OH)–CH_2–CHO
B^-	B	④ CH_3–CH(OH)–CH_2–CHO

The four different products have similar physical properties, making them difficult to separate.

Directed aldol using LDA

On the other hand, LDA is used to remove the α-hydrogen that creates the enolate ion. Because LDA is a strong base, all the carbonyl compound will be converted to the enolate ion. An aldol addition will not be able to occur until the second carbonyl compound is

added. If the second carbonyl compound is added slowly, the chance that it will form an enolate ion and then react with another molecule of its parent compound will be minimized.

```
cyclopentanone  --(LDA)-->  enolate  --(① CH3CH2CH2–CHO added slowly;  ② H2O)-->  2-(1-hydroxybutyl)cyclopentanone

(ring C–CH(OH)–CH2CH2CH3)
```

7. α,β-Unsaturated aldehydes and ketones: nucleophilic addition

The resonance contributors of an α,β-unsaturated carbonyl compound show that the molecule has **two electrophilic sites: the carbonyl carbon and the β-carbon**.

```
  β   α   O                  O⁻                      O⁻
R–CH=CH–C–R   ↔   R–CH=CH–C⁺–R   ↔   R–CH⁺–CH=C–R
```

Thus a nucleophile can add either to the carbonyl carbon or to the β-carbon.

Direct addition (1,2-addition)

Nucleophilic addition to the carbonyl carbon is called direct addition (1,2-addition).

```
Nu⁻  +  R–CH=CH–C(=O)–R   -->   R–CH=CH–C(O⁻)(Nu)–R   --(H3O⁺)-->   R–CH=CH–C(OH)(Nu)–R
```

Conjugate addition (1,4-addition)

Nucleophilic addition to the β-carbon is called conjugate addition or 1,4-addition, because it occurs at the 1 and 4 positions (numbering O = 1, C=O carbon = 2, α-carbon = 3, β-carbon = 4).

```
Nu⁻ + R–CH=CH–C(=O)–R   -->   [ R–CH(Nu)–CH⁻–C(=O)–R  ↔  R–CH(Nu)–CH=C(O⁻)–R ]   (resonance contributors)
        4    3   2 1

   -->   R–CH(Nu)–CH2–C(=O)–R     ⇌     R–CH(Nu)–CH=C(OH)–R
         keto tautomer                   enol tautomer
```

Which one forms?

- **If the nucleophile is a weak base, the conjugate addition product is always formed.** (Margin note in the notes: strong acids have weak conjugate bases.)

```
cyclohexenone  +  ⁻C≡N   --(HCl)-->   3-cyanocyclohexanone       (CN on the
β-carbon of the ring)

CH₂=CH–CO–CH₃  +  HBr   -->   Br–CH₂–CH₂–CO–CH₃

CH₃CH=CH–CO–CH₃  +  (CH₃)₂NH   -->   CH₃–CH(N(CH₃)₂)–CH₂–CO–CH₃
```

- **Strong bases as nucleophiles add directly to the carbonyl carbon.**

```
cyclohexenone  --(1. CH₃MgBr;  2. H₃O⁺)-->   1-methylcyclohex-2-en-1-ol
(OH and CH₃ on the former carbonyl carbon; C=C kept)
```

A second example drawn beside this one is struck out in the notes.

8. Carboxylic acids and their derivatives: carboxylic acids

Carboxylic acids are organic compounds that contain a carbonyl group attached to a hydroxyl group. They are characterised by having one of the following structures:

```
  H–C(=O)–OH        R–C(=O)–OH        Ar–C(=O)–OH
```

Nomenclature

In systematic nomenclature, a carboxylic acid is named by replacing the terminal "e" of the alkane with **"oic acid"**.

Structure	Systematic name	Common name
H–COOH	Methanoic acid	Formic acid
CH_3–COOH	Ethanoic acid	Acetic acid
C_2H_5–COOH	Propanoic acid	Propionic acid
C_4H_9–COOH	Pentanoic acid	Valeric acid
C_5H_{11}–COOH	Hexanoic acid	Caproic acid

Structure	Systematic name	Common name
CH_2=CH–COOH	Propenoic acid	Acrylic acid
C_6H_5–COOH	Benzenecarboxylic acid	Benzoic acid

- In systematic nomenclature, the position of a substituent is designated by a number. The carbonyl carbon is always the C-1 carbon.
- In common nomenclature, the position of a substituent is designated by a lowercase Greek letter, and the carbonyl carbon is not given a designation.

```
CH3-CH2-CH2-CH2-CH2-C(=O)OH          CH3-CH2-CH2-CH2-CH2-C(=O)OH
    6    5    4    3    2    1              ε    δ    γ    β    α
(systematic nomenclature)                  (common nomenclature)
```

Examples:

```
CH3-CH2-CH(OCH3)-COOH      2-methoxybutanoic acid   (α-methoxybutyric acid)
CH3-CH2-CH(Br)-CH2-COOH    3-bromopentanoic acid    (β-bromovaleric acid)
```

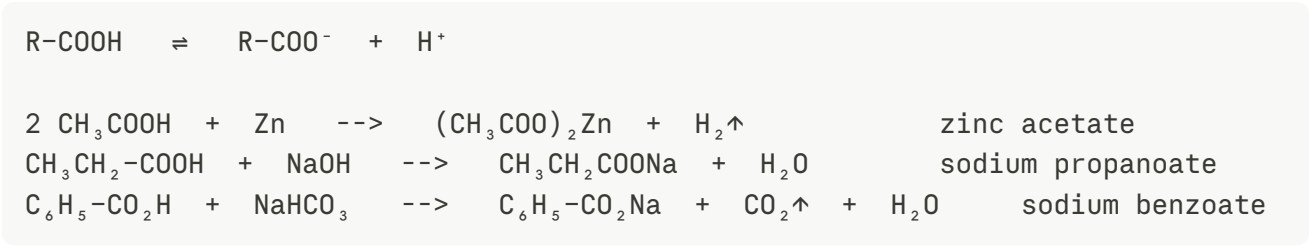
Physical properties

- They are polar compounds and, like alcohols, can form hydrogen bonds with each other and with other kinds of molecules.
- The aliphatic acids show very much the same solubility behaviour as the alcohols: the first 4 members are miscible with water, the 5-carbon acid is partly soluble, and the higher acids are virtually insoluble.
- They have a higher boiling point than alcohols of comparable molecular weight, due to double hydrogen bonding.
- They are characterised by sharp, irritating and unpleasant odours.
- They react with alkali to form salts and water.

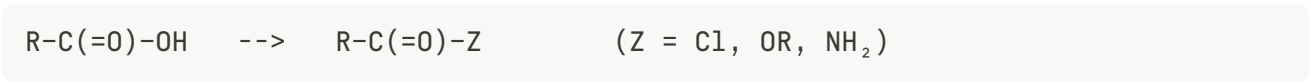
Occurrence and uses: acetic acid is in vinegar and other foods, and is also used in industry as a solvent. Fatty acids come from fats and oils. Benzoic acid is found in drugs and preservatives. Phthalic acid is used to make polyester.

Chemical properties (reactions)

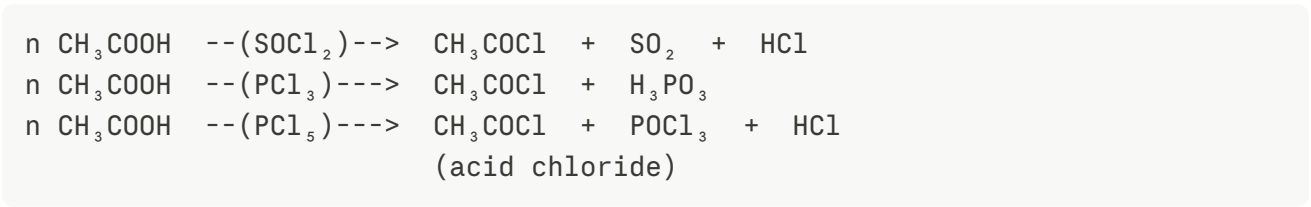
i) Acidity / salt formation



ii) Conversion into functional derivatives

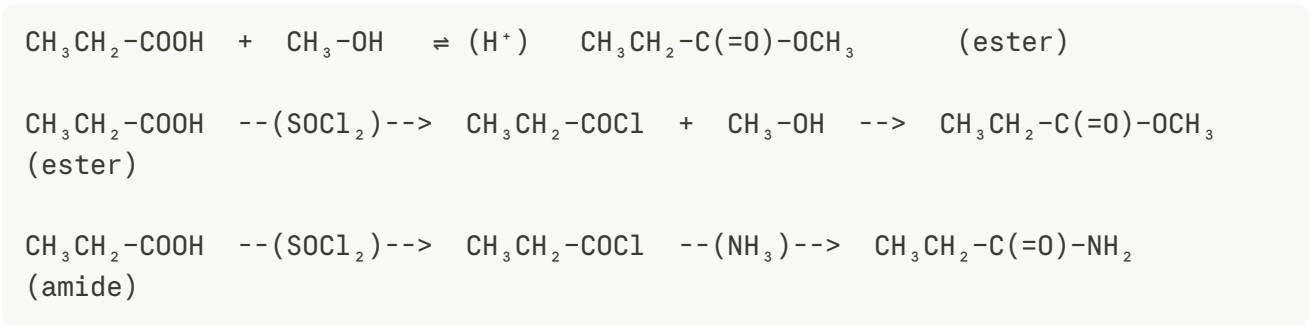


Examples (n = 1, 2 or 3, matching the stoichiometry):



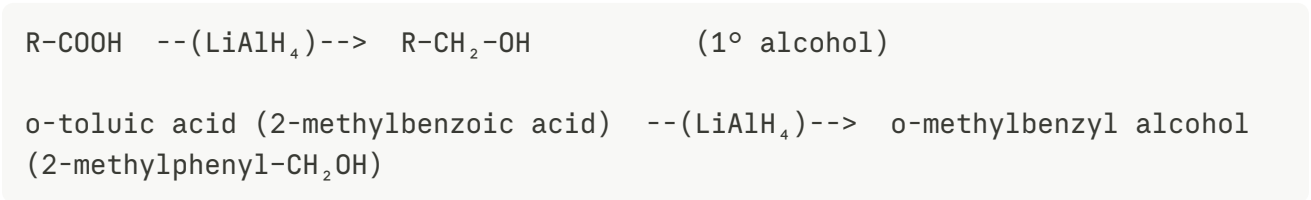
(The PCl_3 line reads "CH_2COCl" in the notes; it should be CH_3COCl.)

Esterification and amide formation



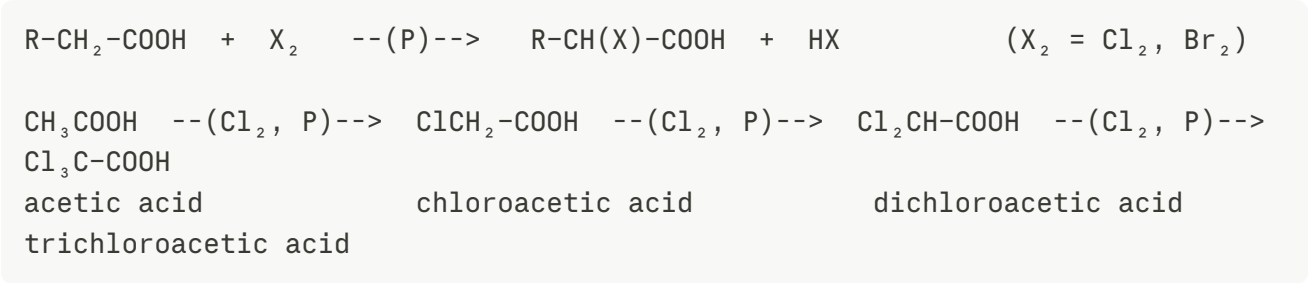
Order of reactivity of the alcohol: CH_3OH > 1° > 2° > 3°.

iii) Reduction



iv) Substitution in the alkyl or aryl group

(a) α-Halogenation of aliphatic acids:

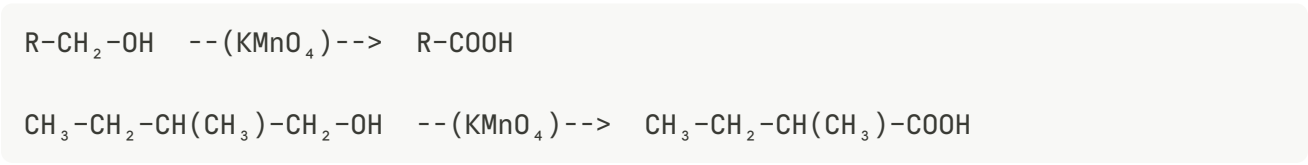


(b) Ring substitution in aromatic acids:

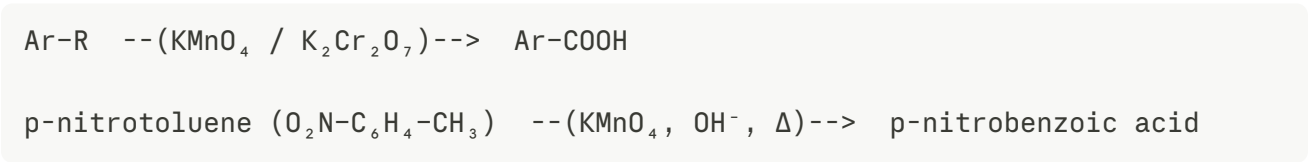


Preparation of carboxylic acids

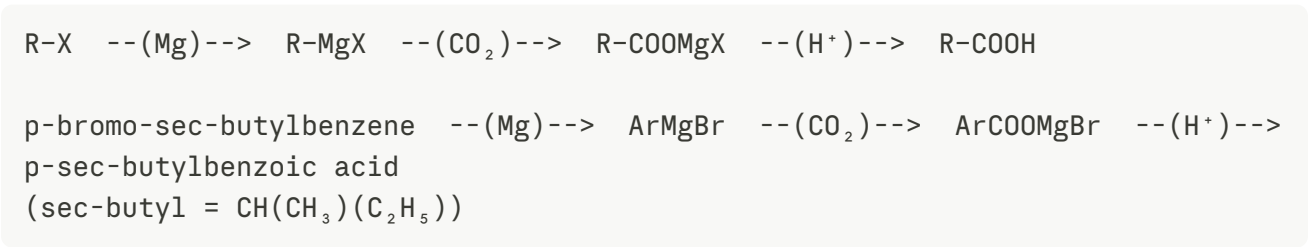
1. Oxidation of 1° alcohols



2. Oxidation of alkylbenzenes



3. Carbonation of Grignard reagents



4. Hydrolysis of nitriles

```
R–C≡N  +  H2O   --(acid or base)-->   R–COOH

benzyl chloride  --(NaCN)-->  C6H5–CH2–CN  --(75% H2SO4, reflux)-->  C6H5–
CH2–COOH
                              phenylacetonitrile
phenylacetic acid

n-butyl chloride  --(NaCN)-->  n-C4H9CN  --(aq. alc. NaOH, reflux)-->  n-
C4H9COO⁻ + NH3  --(H⁺)-->  n-C4H9COOH + NH4⁺
                               n-valeronitrile (pentanenitrile)
valeric acid
```

5. Malonic ester synthesis

```
R–CH(COOC2H5)2   --(H2O, OH⁻, Δ)-->   R–CH(COO⁻)2   --(H⁺)-->   R–CH(COOH)2
--(Δ, 140 °C)-->   R–CH2–COOH  +  CO2
monoalkylmalonic ester
monosubstituted acetic acid
```

6. Special method for phenolic acids (Kolbe reaction)

```
sodium phenoxide (phenol salt)  +  CO2   --(125 °C, 4–7 atm)-->   sodium
salicylate (o-HO–C6H4–COONa)   --(H⁺)-->   salicylic acid (o-HO–C6H4–COOH)
```

9. Acid chlorides and acid anhydrides

Acid chlorides

Preparation: they are prepared from the corresponding acids by reaction with thionyl chloride, phosphorus trichloride or phosphorus pentachloride, as discussed in section 8.

Reactions

1. Conversion into acids and derivatives

```
R–C(=O)–Cl  +  H–Z   -->   R–C(=O)–Z  +  HCl          (written with "H2" in
the notes; read as H–Z)
```

(a) Conversion into acids (hydrolysis):

```
R–C(=O)–Cl  +  H2O   -->   R–C(=O)–OH  +  HCl
```

(b) Conversion into amides (ammonolysis):

```
R–C(=O)–Cl  +  2 NH3   -->   R–C(=O)–NH2  +  NH4Cl
```

(c) Conversion into esters:

```
R–C(=O)–Cl  +  R'–O–H   -->   R–C(=O)–OR'  +  HCl
```

2. Formation of ketones

(a) Friedel–Crafts acylation:

```
R–C(=O)–Cl  +  Ar–H   --(AlCl3 or other Lewis acid)-->   R–C(=O)–Ar  +  HCl
```

(b) Reaction with organocopper compounds:

```
R'–Li  --(CuX)-->  R'–CuLi (lithium organocuprate)

R'–CuLi  +  R–C(=O)–Cl  (or Ar–C(=O)–Cl)   -->   R–C(=O)–R'  (or Ar–C(=O)–R')
```

3. Formation of aldehydes by reduction

```
R–C(=O)–Cl  (or Ar–C(=O)–Cl)   --(LiAlH(O-t-Bu)3)-->   R–CHO  (or Ar–CHO)
```

Acid anhydrides

Preparation

(i) From acetic acid via ketene:

```
CH3COOH  --(AlPO4, 700 °C)-->  H2O  +  CH2=C=O  --(CH3COOH)-->  CH3–C(=O)–O–
C(=O)–CH3
(acetic acid)                        (ketene)                     (acetic
anhydride)
```

(ii) From dicarboxylic acids:

```
HOOC–CH2–CH2–COOH   --(heat)-->   succinic anhydride (five-membered ring O=C–
CH2–CH2–C=O bridged by O)  +  H2O
(dicarboxylic acid)
```

Reactions

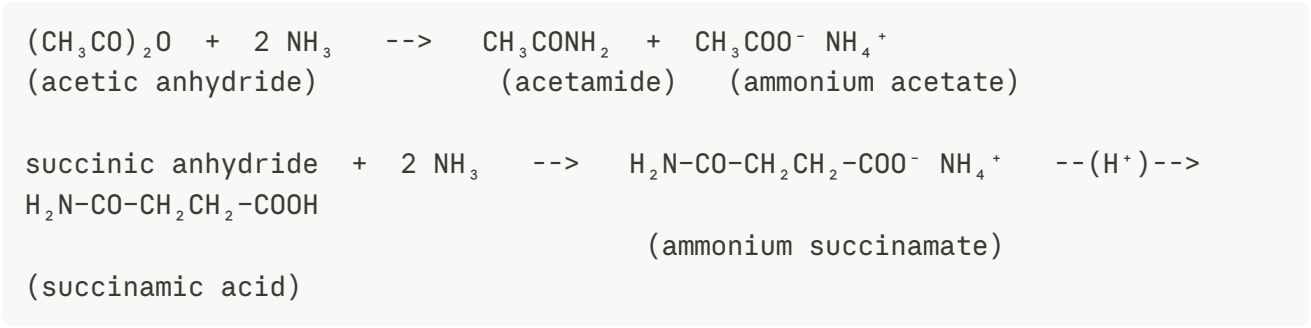
(i) Conversion into acids and acid derivatives



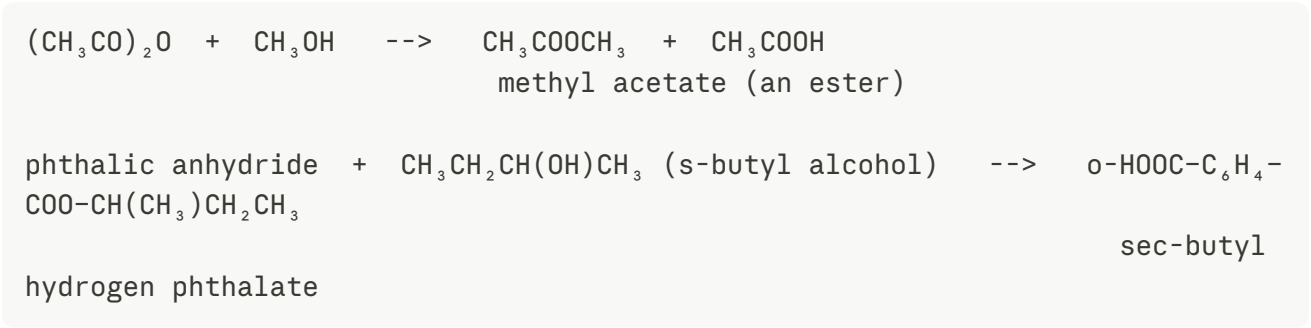
(a) Conversion into acids (hydrolysis):



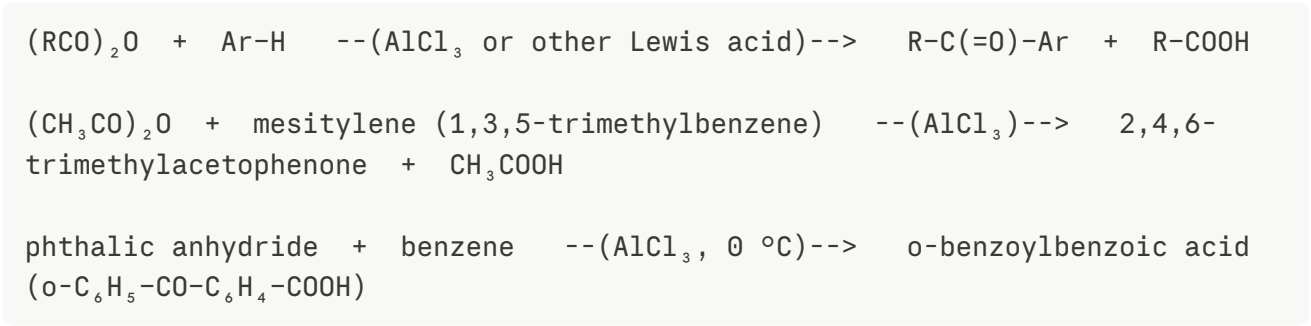
(b) Conversion into amides (ammonolysis):



(c) Conversion into esters (alcoholysis):



(ii) Formation of ketones (Friedel–Crafts acylation)



10. Amides and esters

Amides

Preparation

```
R–C(=O)–Cl  +  NH3   -->   R–C(=O)–NH2
```

Reactions

i) Hydrolysis

```
R–C(=O)–NH2  +  H2O   --(H+)-->   R–C(=O)–OH  +  NH4+
                      --(OH-)-->  R–COO-  +  NH3

benzamide (C6H5–CONH2)  +  H2SO4  +  H2O   -->   C6H5–COOH  +  NH4+ HSO4-
CH3CH2–CONH2  +  NaOH  +  H2O   -->   CH3CH2–COONa  +  NH3
```

ii) Conversion into imides

```
phthalic anhydride  --(2 NH3)-->  o-C6H4(CONH2)(COO- NH4+)  --(H+)-->  o-
C6H4(CONH2)(COOH)

phthalic anhydride  --(NH3, Δ)-------------------------------------->
phthalimide
o-C6H4(CONH2)(COO- NH4+)  --(Δ, 300 °C)-------------------------->
phthalimide
o-C6H4(CONH2)(COOH)  --(Δ)---------------------------------------->
phthalimide

phthalimide = benzene ring fused to a five-membered ring: C(=O)–NH–C(=O)
```

iii) Hofmann degradation

```
R–CONH2  (or Ar–CONH2)   --(OBr-)-->   R–NH2  (or Ar–NH2)  +  CO3 2-
```

Esters

Preparation

- **From acids** (reactivity of R′OH: 1° > 2° > 3°):

```
R–C(=O)–OH  +  R'O–H   ⇌ (H+)   R–C(=O)–OR'  +  H2O
```

- **From acid chlorides or anhydrides:**

```
R–C(=O)–Cl  +  R'–OH   -->   R–C(=O)–OR'  +  HCl
(RCO)2O  +  R'–OH   -->   R–COOR'  +  R–COOH
```

- **From esters (transesterification):**

```
R–C(=O)–OR'  +  R''O–H   ⇌ (H⁺ or ⁻OR'')   R–C(=O)–OR''  +  R'–OH
```

Mechanism of acid-catalysed transesterification (the notes title this "acid-catalyzed esterification")

1. Protonation of the carbonyl oxygen of Ester A, R–C(=O)–OR′, gives R–C(=O⁺H)–OR′ ↔ R–C⁺(OH)–OR′.
2. Alcohol B (R″–OH) attacks the carbonyl carbon: R–C(OH)(OR′)–O⁺(H)R″.
3. Proton transfer gives R–C(OH)(O⁺HR′)–OR″.
4. Loss of Alcohol A (R′–OH) gives R–C(=O⁺H)–OR″.
5. Loss of H⁺ gives **Ester B**, R–C(=O)–OR″. Each step is reversible.

Mechanism of base-catalysed transesterification

```
R–C(=O)–OR'  +  ⁻OR''   ⇌   R–C(O⁻)(OR')(OR'')   ⇌   R–C(=O)–OR''  +  ⁻OR'
Ester A         alkoxide B     tetrahedral intermediate    Ester B
alkoxide A
```

Reactions of esters

Conversion into acids and derivatives

(a) Conversion into acids (hydrolysis):

```
R–C(=O)–OR'  +  H2O   --(H⁺)-->   R–C(=O)–OH  +  R'–OH
                      --(OH⁻)-->  R–COO⁻  +  R'–OH
```

(b) Conversion into amides:

```
R–C(=O)–OR'  +  NH3   -->   R–C(=O)–NH2  +  R'–OH
```

(c) Conversion into esters (transesterification):

```
R–C(=O)–OR'  +  R''–OH   ⇌ (acid or base)   R–C(=O)–OR''  +  R'–OH

glyceride (a fat):  CH2–O–C(=O)–R / CH–O–C(=O)–R' / CH2–O–C(=O)–R''   +
CH3OH   --(acid or base)-->
    R–COOCH3 + R'–COOCH3 + R''–COOCH3  (mixture of methyl esters)   +
HOCH2–CH(OH)–CH2OH  (glycerol)
```

Reaction with Grignard reagents

```
R–C(=O)–OR'  +  2 R''–MgX   -->   R–C(OH)(R'')₂        (3° alcohol)
```

Reduction to alcohols

- Catalytic hydrogenation (hydrogenolysis):

```
R–COOR'  +  2 H₂   --(CuO·CuCr₂O₄, 250 °C, 3000–6000 lb/in²)-->   R–CH₂OH  +
R'–OH
```

- Chemical reduction:

```
4 R–COOR'  +  2 LiAlH₄   --(anhydrous ether)-->   { LiAl(OCH₂R)₄  +
LiAl(OR')₄ }   --(H⁺)-->   { R–CH₂OH,  R'OH }
```

Reaction with carbanions: Claisen condensation

```
CH₃–C(=O)–OR'  +  (CH₃)₂CH–C(=O)–OR''   --(⁻OC₂H₅)-->   CH₃–C(=O)–C(CH₃)₂–
C(=O)–OR''
                                                          (a β-keto ester)
```